Progress Note: 2-Period Igami Model with Agglomeration

14.273 Mini Project

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1 Overview

This note documents the first extension of Igami (2017) to incorporate **innovation spillovers / agglomeration effects** in a simplified 2-period setting. The goal is to understand whether these effects undo the innovator's dilemma found by Igami.

2 Model

2.1 State and Firm Types

The market state at each period t is $s_t = (n_o, n_b, n_n, n_{pe})$, where:

- n_o: old incumbents (have *not* adopted new technology),
- n_b: "both" incumbents (have adopted; produce with new tech),
- n_n: new entrants (using new technology),
- n_{pe}: potential entrants.

Active firms in the product market are those with $n_o + n_b + n_n > 0$. "Both" firms are multi-product: they continue to sell old-gen (at cost c_o) while also selling new-gen (at cost $c_{n,t}$). This is the source of the **cannibalization effect**: by expanding new-gen supply, a "both" firm depresses the old-gen price it receives, creating a disincentive to innovate that is absent for pure new entrants.

2.2 Timing Within Each Period

1. **Product market:** Cournot competition in *two differentiated markets* (old-generation and new-generation products) with linear inverse demand:

$$P_o = A - \frac{B}{M} Q_o - \frac{\rho B}{M} Q_n, \tag{1}$$

$$P_n = A - \frac{B}{M} Q_n - \frac{\rho B}{M} Q_o, \tag{2}$$

   where $\rho \in [0, 1)$ is the cross-market substitution parameter. Old firms (n_o) compete only in the old-gen market at cost c_o. New entrants (n_n) compete only in the new-gen market at cost $c_{n,t}$. "Both" firms (n_b) are *multi-product*: they sell in both markets simultaneously and internalize the **cannibalization** effect — their new-gen output reduces demand for their own old-gen product.

2. **Private shocks:** Each firm draws i.i.d. cost shocks $\varepsilon_{i,a} \sim \text{EVT1}(0, \sigma)$ for each action a.

3. **Discrete choices** (sequential order: old $\to$ both $\to$ new $\to$ pe):

   - old: $\{\text{stay}, \text{innovate}, \text{exit}\}$

- both/new: $\{\text{stay}, \text{exit}\}$
- pe: $\{\text{enter}, \text{quit}\}$

Each type observes the *realized* outcomes of all earlier movers before making its own decision.

4. **Transition** to period $t+1$.

2.3 Agglomeration Extension

The new-technology marginal cost depends on the number of firms already using it:

$$c_{n,t} = c_{n,0} - \gamma \cdot (n_{b,t} + n_{n,t}), \qquad \gamma \geq 0. \tag{3}$$

When $\gamma = 0$ the model reduces to the Igami baseline. When $\gamma > 0$, a larger innovative cluster lowers costs for all new-tech firms (knowledge spillover / agglomeration effect).

2.4 Value Functions and CCPs

Because firms move sequentially within each period and each type observes the *realized* outcomes of all earlier movers, each type's decision problem is effectively a *single-agent problem*: earlier-mover uncertainty has been resolved, and later-mover choices enter only through pre-computed CCPs. Combined with i.i.d. EVT1 private cost shocks, this yields closed-form logit CCPs at each stage:

$$P(a \mid \tau, \tilde{s}) = \frac{\exp\big(\bar{V}_a(\tau, \tilde{s})/\sigma\big)}{\sum_{a'} \exp\big(\bar{V}_{a'}(\tau, \tilde{s})/\sigma\big)}, \tag{4}$$

where $\tilde{s}$ is the *realized intermediate state* after all earlier movers have acted, and $\bar{V}_a$ is the systematic utility:

$$\bar{V}_a(\tau, \tilde{s}) = \pi_\tau(s_0) + \text{cost}(a) + \beta\, \mathbb{E}[V_{t+1}(\tau', s') \mid \tilde{s},\, a_i = a,\, \text{CCP}_{\text{later}}]\,. \tag{5}$$

Here $\text{CCP}_{\text{later}}$ denotes pre-computed CCPs of same-type peers and all later-moving types; flow profits $\pi_\tau(s_0)$ are evaluated at the period-opening state.

2.5 Solution Method

The model is solved by backward induction, following Igami (2017):

1. **Terminal period ($t = 1$):** $V_1(\tau, s_1) = \pi_\tau(s_1)$, where Cournot profits use the agglomeration-adjusted cost $c_{n,1} = c_{n,0} - \gamma(n_{b,1} + n_{n,1})$.

2. **Period $t = 0$:** For each state s_0, stages are solved in *reverse move order* (PE $\to$ new $\to$ both $\to$ old), exploiting the sequential structure:

   - **Stage 4 (PE):** Each PE firm observes the full realized intermediate state $\tilde{s}$ and solves a binary logit $\{\text{enter}, \text{quit}\}$. We compute CCPs for all intermediate states reachable from s_0.

- **Stage 3 (new):** Each new firm solves a binary logit $\{\text{stay}, \text{exit}\}$, conditioning on pre-computed PE CCPs to integrate over the PE-stage continuation value.
- **Stage 2 (both):** Each both firm solves a binary logit, conditioning on new + PE CCPs. Innovators locked in from Stage 1 (k_{io} old firms that chose innovate) are added to n_b before this stage.
- **Stage 1 (old):** Each old firm solves a multinomial logit $\{\text{stay}, \text{innovate}, \text{exit}\}$, conditioning on all lower-stage CCPs.

No fixed-point iteration is needed within the period: the sequential structure makes each stage a closed-form single-agent problem given pre-computed downstream CCPs. Marginal CCPs for later-moving types are recovered by integrating over the equilibrium transition distribution (products of independent Binomial/Multinomial draws).

3 Calibration

We summarize the parameters we use for our calibration and later to simulate the data in Table 1.

Table 1: Parameter values for calibration and simulation

Parameter	Description	Value
A	Demand intercept	3.0
B	Demand slope	1.0
M	Market size	1.0
c_o	Old-tech marginal cost	1.5
$c_{n,0}$	New-tech baseline cost	0.5
β	Discount factor	0.9
κ	Innovation cost	0.3
ϕ	Entry cost	0.2
σ	EVT1 scale	1.0
γ	Agglomeration (comparative statics)	varied
ρ	Cross-market substitution	0.5

The initial state of interest is $s_0 = (n_o, n_b, n_n, n_{pe}) = (4, 1, 1, 2)$, loosely inspired by the early HDD market structure in Igami (2017).

4 Results

4.1 Baseline vs. Agglomeration

Table 2 compares equilibrium CCPs at s_0 for $\gamma = 0$ (baseline) and $\gamma = 0.05$.

Table 2: Baseline vs. agglomeration

	$\gamma = 0$	$\gamma = 0.05$
$P(\text{innovate} \mid \text{old})$	0.3135	0.3206
$P(\text{enter} \mid \text{pe})$	0.5061	0.5139

Notes: Equilibrium CCPs at $s_0 = (4, 1, 1, 2)$ for the baseline ($\gamma = 0$) and agglomeration ($\gamma = 0.05$) specifications.

Agglomeration raises both the innovation and entry rates. Intuitively, a larger innovative cluster lowers future costs for new-tech firms, increasing the expected continuation value of innovating or entering.

4.2 Comparative Statics in γ

Table 3 reports equilibrium CCPs as γ varies from 0 to 0.30, and Figure 1 plots the same.

Table 3: Comparative statics in γ

γ	$P(\text{innovate}\|\text{old})$	$P(\text{enter}\|\text{pe})$
0.00	0.3135	0.5061
0.05	0.3206	0.5139
0.10	0.3282	0.5221
0.15	0.3314	0.5258
0.20	0.3318	0.5263
0.25	0.3315	0.5260
0.30	0.3316	0.5260

Notes: Equilibrium CCPs at $s_0 = (4, 1, 1, 2)$ as the agglomeration parameter γ varies from 0 to 0.30.

4.3 Comparative Statics in ρ (Cannibalization)

Figure 2 plots equilibrium CCPs as the substitution parameter ρ varies from 0 (independent markets) to 0.95 (near-homogeneous goods), holding $\gamma = 0.05$.

4.4 Discussion

Three features of the comparative statics are notable:

1. **Positive agglomeration effect:** Higher γ unambiguously raises the innovation rate for small values of γ. This confirms the model's central prediction: knowledge spillovers encourage adoption.

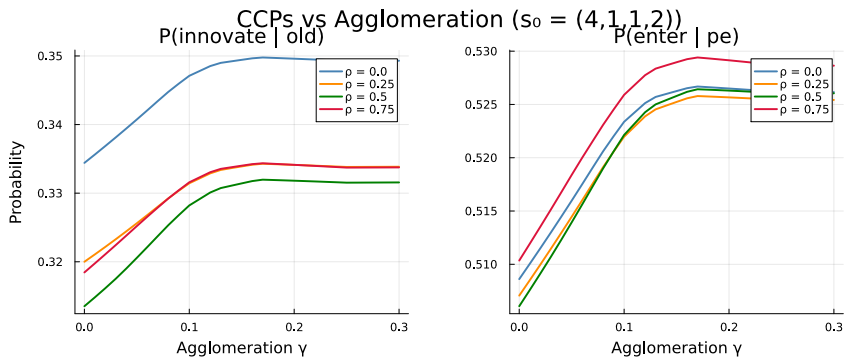

Figure 1: $P(\text{innovate} \mid \text{old})$ and $P(\text{enter} \mid \text{pe})$ as functions of γ at $s_0 = (4, 1, 1, 2)$.

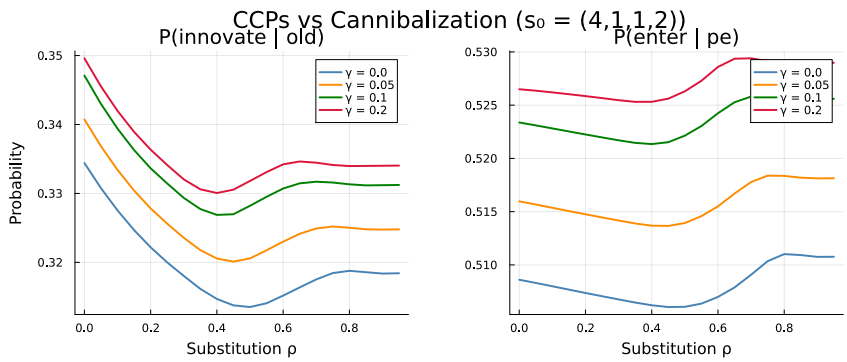

Figure 2: $P(\text{innovate} \mid \text{old})$ and $P(\text{enter} \mid \text{pe})$ as functions of ρ at $s_0 = (4, 1, 1, 2)$, $\gamma = 0.05$.

2. **Cannibalization deters innovation:** Higher ρ (stronger substitution between old-gen and new-gen) reduces the profit of "both" firms by depressing old-gen prices. In the calibration, per-firm "both" profit falls from ≈ 0.531 at $\rho = 0$ to ≈ 0.327 at $\rho = 0.5$. This is the central mechanism in Igami (2017): incumbents are reluctant to innovate because doing so cannibalizes their existing old-gen revenue. However, the relation is nonmonotone: as ρ approaches 1, the two markets become nearly homogeneous and the incumbent benefits from innovating because it can sell more of the new-gen product.

3. **Saturation:** The innovation rate saturates near 1/3 for large γ. This is a feature of the EVT1 / logit structure: with large σ relative to payoff differences, all actions approach equal probability. In a richer calibration (smaller σ), the effect would be more pronounced.

5 Regional Extension

5.1 Regional Structure

To study the interaction between industrial policy and agglomeration we extend the state space with a **region** attribute. There are $R = 3$ regions (labelled $r \in \{1, 2, 3\}$); each firm is born into a region and never leaves it, and potential entrants sit in per-region pools $n_{pe,r}$.

Product-market competition remains **global** — all active firms meet in the same two-market Cournot game of equations (1)–(2) — but agglomeration spillovers are now **local**:

$$c_{n,r,t} = c_{n,0} - \gamma_r \cdot \big(n_{b,r,t} + n_{n,r,t}\big), \qquad r = 1, \ldots, R. \tag{6}$$

Because $c_{n,r,t}$ varies across regions, firms of type "both" and "new" face *heterogeneous* marginal costs in the new-gen market. The Cournot stage therefore loses symmetry across regions, and equilibrium quantities are obtained by solving an asymmetric FOC system of up to $4R$ equations per state. Within-region slots remain symmetric (all firms in the same $(type, r)$ cell pick the same quantity), so per-firm profits are still characterised by $\pi_{o,r} = bq_{oo,r}^2$, $\pi_{n,r} = bq_{nn,r}^2$, and $\pi_{b,r} = b(q_{bo,r}^2 + q_{bn,r}^2) + 2s\, q_{bo,r} q_{bn,r}$ where $b = B/M$ and $s = \rho B/M$.

Each firm privately captures the lower cost when it innovates but does not internalize the benefit it confers on regional peers — a standard **local agglomeration externality** that creates a wedge between the social and private return to innovation and opens the door to regionally targeted industrial policy.

5.2 Solution with Regional States

The sequential-move structure (old → both → new → pe) carries over unchanged, and we extend the same idea one level down: **within each stage, regions move sequentially**, $r = 1 \to 2 \to 3$. When firms in region r decide, they observe the realized outcome of regions $1, \ldots, r-1$ and integrate over the (yet-to-play) strategies of regions $r+1, \ldots, R$. This is the regional analogue of Igami's cross-type sequencing: it makes every sub-stage a single-agent problem whose CCP is a *scalar* fixed point rather than a joint fixed point across regions, and it removes any coupling in the solver between regions at the same stage.

Concretely, we solve each stage in reverse region order. Region R's scalar FP is computed first for every reachable sub-state, and cached. Region $R-1$'s FP is then computed given the cached region-R continuation, and so on down to region 1. Across stages, the usual layered caches still apply: stage-4 (PE) depends only on the post-new-stage state and is reused across initial states via $\mathrm{EV}_{\text{after pe}}[s_{an}, r]$; stages 3, 2, 1 cascade through $\mathrm{EV}_{\text{after new}}$, $\mathrm{EV}_{\text{after both}}$, and $\mathrm{EV}_{\text{after old}}$, each keyed by (State, r). Because each sub-state's CCP is a function of the realized history, the per-region CCPs reported in the results below are *marginalised* over the equilibrium play of earlier-moving regions, computed by a forward traversal of the sub-stage tree.

Because the state space now has $4R = 12$ integer coordinates, we cap $N_{\max} = 6$ on this branch (about 1.8×10^4 enumerated states) and pick initial states with total firm count ≤ 6. One visible artefact of the sequential-move convention is a small first-mover asymmetry in the baseline: with uniform $\gamma_r = 0.05$ the reported innovation CCPs are $(0.3333, 0.3330, 0.3326)$ rather than the exactly-symmetric values one would get under simultaneous moves within a stage. Region 1 commits without observing anyone, while region 3 best-responds to the realised history; the gap is $\leq 10^{-3}$ in all experiments and does not affect any of the counterfactual comparisons.

5.3 Baseline and comparative statics

We solve the regional model at two initial states, both with total firm count 6:

- **Symmetric s_0:** $s_0^{\text{sym}} = \big((1,1,1),(1,1,1),(0,0,0),(0,0,0)\big)$ — one old incumbent and one "both" incumbent in each region, no entrants.

- **Clustered s_0:** $s_0^{\text{clust}} = \big((1,1,1),(3,0,0),(0,0,0),(0,0,0)\big)$ — same number of firms of each type, but all three "both" incumbents are concentrated in region 1.

Starting from s_0^{sym} with uniform $\gamma_r = 0.05$ we verify a **symmetry check**: the three regions' CCPs are nearly identical, with only tiny differences (on the order of 10^{-3}) that reflect the sequential region-move structure discussed above rather than any substantive break in the model's permutation symmetry when the state and parameters are symmetric.

Two comparative statics are then run against this baseline. **Comparative Static A** subsidises innovation in region 1 by tripling its agglomeration strength to $\gamma_1 = 0.15$ while leaving $\gamma_2 = \gamma_3 = 0.05$. **Comparative Static B** holds γ_r uniform but starts from the clustered initial state s_0^{clust}, asking whether a historical concentration of innovators is enough to lock in innovation leadership even without a policy tilt.

Table 4: Regional innovation CCPs by scenario

Scenario	$P(\text{innov}\|o, r{=}1)$	$P(\text{innov}\|o, r{=}2)$	$P(\text{innov}\|o, r{=}3)$
Baseline	0.3333	0.3330	0.3326
CS-A: subsidy r1	0.3594	0.3263	0.3260
CS-B: cluster r1	0.3374	0.3236	0.3234

Notes: Reports $P(\text{innovate} \mid \text{old}, r)$ for each region. Baseline uses s_0^{sym} with uniform $\gamma = 0.05$; CS-A sets $\gamma_1 = 0.15$ (regions 2 and 3 held at 0.05); CS-B uses the clustered initial state s_0^{clust} with uniform $\gamma = 0.05$.

5.4 Discussion

Three findings stand out in Table 4 and Figure 3:

1. **Targeted subsidies work, but leak.** Raising γ_1 from 0.05 to 0.15 moves region 1's innovation probability from 0.3333 to 0.3594, a 0.3333-to-0.3594 gain of roughly two and a half percentage points. Regions 2 and 3 fall from 0.3330 to 0.3263— a small but clearly negative spillover driven entirely by the *global* product market: more innovation in region 1 lowers expected new-gen prices everywhere, eroding the option value of innovating in the other regions.

2. **Historical clusters alone are weak.** Starting from s_0^{clust} the incumbent cluster in region 1 barely moves innovation there (0.3374 vs. 0.3333 in the symmetric baseline): with only one remaining old firm in region 1, the region's marginal old firm is already close to saturating its own local spillover. The larger effect shows up in regions 2 and 3, whose CCPs fall to 0.3236 because of the global-market pressure from the concentrated region-1 innovators. A one-shot historical endowment is therefore a much weaker policy lever than a continuing subsidy.

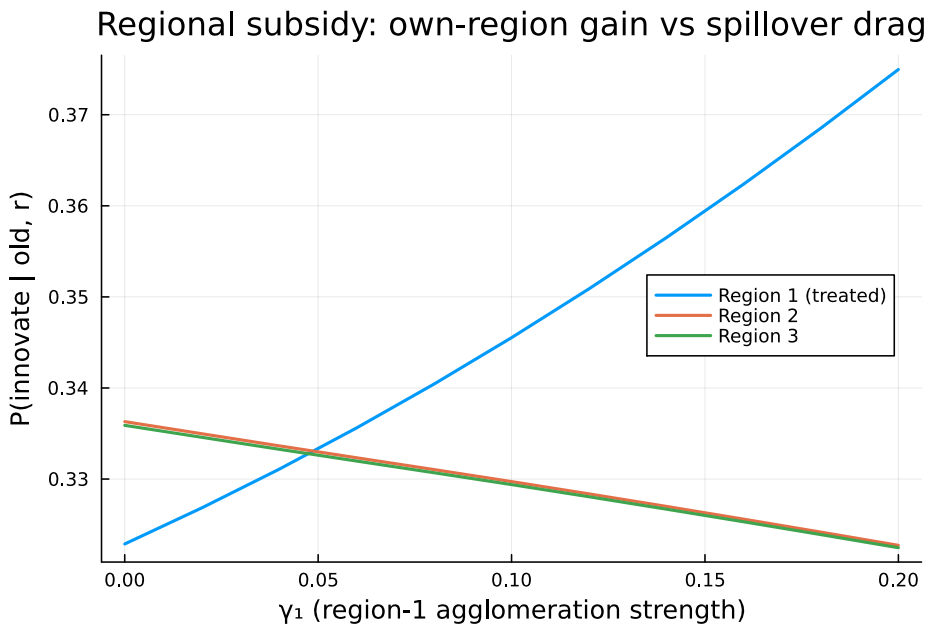

Figure 3: Regional innovation rates as γ_1 varies while $\gamma_2 = \gamma_3 = 0.05$ is held fixed. Region 1 responds positively to its own subsidy; regions 2 and 3 are slightly depressed through the global Cournot channel.

3. **The spillover wedge is still local.** Comparing CF-A and CF-B makes clear that what drives region 1's innovation is not how many innovators are in the region today but the *marginal reduction in new-tech cost* each additional innovator delivers. The policy CF-A tilts that reduction directly, which is why it dominates a pure history-based CF-B at the same total cluster size.

The two experiments illustrate how the regional model bridges Igami's replacement-effect story with the industrial-policy literature: even a modest, regionally targeted spillover subsidy reshapes the geographic distribution of innovation at the cost of a small drag on the rest of the world, with magnitudes that depend on the level of cannibalization ρ and the strength of the local externality γ_r.

6 Estimation on Simulated Data

As a first pass at recovering the model's structural parameters from data, we simulate a firm-level dataset from the regional model under the calibration of Table 1 and then estimate (γ, κ, ϕ), treating the demand parameters, the EVT1 scale σ, the cost $c_{n,0}$, and the state cap $N_{\max}$ as known.

6.1 Simulated dataset

The simulator (`code/scripts/simulate_data.jl`) draws 500 independent markets from the 2-period regional model. Each market gets its own initial state $s_0 \sim \text{unif}\{0, 1, 2\}^{4R}$ (rejecting

empty or over-capacity draws), so the panel covers a range of local cluster sizes. For every firm in every period we record region, firm type, the two Cournot quantities, realised profit, and—for period 1 only—the discrete action. The structural parameters used to draw the data are deliberately *not* stored in the CSV.

6.2 Two-step estimator

Because equilibrium quantities are a deterministic function of the state and the spillover vector γ, identification splits cleanly: γ is pinned by the Cournot block and (κ, ϕ) by the discrete actions, conditional on γ. We exploit this with a two-step estimator (see `notes/estimation.md` for the full writeup).

Step 1 — NLS for γ. For each market we reconstruct s_0 and s_1 by counting firms of each type and region in the corresponding period. We then solve

$$\widehat{\gamma} \;=\; \arg\min_{\gamma} \sum_{m,t,r,\,\mathrm{slot}} \big(q^{\mathrm{obs}}_{m,t,r,\mathrm{slot}} - \hat{q}_{m,t,r,\mathrm{slot}}(\gamma)\big)^2, \tag{7}$$

starting from $\gamma^0 = (0, 0, 0)$ and leaving γ *unconstrained*. Because the simulated quantities are noiseless, the minimum is machine zero (SSR $= 1.34e-13$) and $\widehat{\gamma}$ equals the truth to all reported digits; this step doubles as a sanity check that state reconstruction is correct.

Step 2 — MLE for (κ, ϕ). Holding $\gamma = \widehat{\gamma}$ fixed, we maximise the period-1 action log-likelihood over (κ, ϕ). The likelihood *replays* the sequential-move order used by the solver: within each stage, regions act in order $r = 1 \to 2 \to 3$, and each firm's contribution uses the stage CCP evaluated at the sub-state its firm actually faced (not the marginal CCP integrated over the forward traversal). Per market,

$$\begin{aligned}
\ell_m(\kappa, \phi) = &\sum_{r=1}^{R} \Big[k_r^{\mathrm{so}} \log p_{\mathrm{so}}(\tilde{s}_r^{\mathrm{OLD}}) + k_r^{\mathrm{io}} \log p_{\mathrm{io}}(\tilde{s}_r^{\mathrm{OLD}}) + k_r^{\mathrm{eo}} \log\big(1 - p_{\mathrm{so}} - p_{\mathrm{io}}\big)\Big] \\
&+ \sum_{r=1}^{R} \Big[k_r^{\mathrm{sb}} \log p_{\mathrm{sb}}(\tilde{s}_r^{\mathrm{BOTH}}) + k_r^{\mathrm{eb}} \log\big(1 - p_{\mathrm{sb}}\big)\Big] \\
&+ \sum_{r=1}^{R} \Big[k_r^{\mathrm{sn}} \log p_{\mathrm{sn}}(\tilde{s}_r^{\mathrm{NEW}}) + k_r^{\mathrm{en}} \log\big(1 - p_{\mathrm{sn}}\big)\Big] \\
&+ \sum_{r=1}^{R} \Big[k_r^{\mathrm{ep}} \log p_{\mathrm{ep}}(\tilde{s}_r^{\mathrm{PE}}) + k_r^{\mathrm{op}} \log\big(1 - p_{\mathrm{ep}}\big)\Big],
\end{aligned} \tag{8}$$

where the $k_r^{\bullet}$ are observed action counts in region r and the $\tilde{s}_r^{\bullet}$ are the sub-states produced by sequentially updating s_0 with the realised outcomes of earlier regions and stages. The total objective is $\ell(\kappa, \phi) = \sum_m \ell_m(\kappa, \phi)$, maximised by Nelder–Mead starting from $(\kappa^0, \phi^0) = (1.0, 1.0)$ — intentionally far from the calibration so the optimiser must traverse the parameter space. Standard errors come from the BHHH estimator $V = (\sum_m g_m g_m')^{-1}$ with central finite-difference gradients.

6.3 Results

Table 5 reports the estimates against the data-generating parameters. The two-step estimator recovers $\widehat{\gamma} = (0.0500, 0.0500, 0.0500)$ exactly (Step 1), and the action MLE delivers $\widehat{\kappa} = 0.2849$ (s.e. 0.0815, truth 0.3000) and $\widehat{\phi} = 0.1635$ (s.e. 0.0792, truth 0.2000). Both κ and ϕ lie within roughly one standard error of their true values; given 500 markets and a handful of deciders per market, this is about the precision the sample supports. The period-1 action log-likelihood at the optimum is -2079.207.

Table 5: Two-step estimation of (γ, κ, ϕ)

Parameter	Truth	Estimate	S.E.
κ	0.3000	0.2849	0.0815
ϕ	0.2000	0.1635	0.0792
γ_1	0.0500	0.0500	—
γ_2	0.0500	0.0500	—
γ_3	0.0500	0.0500	—

Notes: Estimates from 500 simulated markets. Step 1 identifies γ from Cournot quantities; Step 2 estimates (κ, ϕ) by period-1 action MLE conditional on $\widehat{\gamma}$. Standard errors (BHHH) are reported only for the MLE step.

7 Next Steps: Counterfactuals

1. **Bilateral spillover pools.** Allow neighbouring regions to share an enlarged effective innovator pool (e.g. regions 1 and 2 pool their N^{inn}), and trace how cross-regional knowledge flows reshape the leakage pattern documented in Table 4.

2. **Welfare and industrial policy.** Compute consumer surplus, producer surplus, and fiscal cost under the region-1 subsidy of Comparative Static A, and report the net welfare change to assess whether targeted R&D subsidies are justified once the global Cournot spillover is priced in.